

Technology Stack

Date	07 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	2 Marks

Technology Stack Details :

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Bootstrap, Chart.js, AOS Animation, Lottie	Used to create a responsive, interactive, and user-friendly interface with modern animations, charts, and mobile compatibility.
2	Backend / Server-Side	Python 3.10, Flask Framework	Flask is a lightweight Python framework used to integrate the Machine Learning model, process user requests, and serve dynamic web pages efficiently.
3	Database / Data Storage	CSV Dataset, Pickle (.pkl) Model Files	The dataset is stored in CSV format for training, while the trained Random Forest model and scaler are stored as Pickle files for fast predictions.
4	Cloud / Hosting / Deployment	Render, Docker	Render is used to deploy and host the Flask application with automatic deployment from GitHub. Docker provides a consistent and portable deployment environment.
5	Version Control & CI/CD	Git, GitHub	Git is used for version control, and GitHub is used for source code management, collaboration, and automatic deployment to Render.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, OpenPyXL, VS Code, Jupyter Notebook	These tools are used for data preprocessing, Machine Learning model development, visualization, analysis, report generation, and application development.

Technology Summary

Smart Lender AI is developed using Python and the Flask framework. The frontend is built with HTML5, CSS3, JavaScript, Bootstrap, Chart.js, AOS Animation, and Lottie to provide an interactive user experience. The Machine Learning model is developed using Scikit-learn with Pandas and NumPy for data preprocessing. The trained model is stored as Pickle files and the application is version-controlled using Git and GitHub. Finally, the application is deployed on Render with Docker support, ensuring reliable, scalable, and cloud-based access.